Quiz
Test yourself 1

TEST YOURSELF 1

ANSWERS ON P. 564

1.01

1 Convert each statement to an algebraic expression.

a the cost of n pizzas at \$$P$ each

b how many 3s divide into m

c 5 rides at \$$A$ each and 8 rides at \$$B$ each

d decrease \$100 by \$$y$

e the number of minutes in H hours

f the number of litres in m mL

g multiply P by 4 and add 5

h add 5 to T, then multiply by 4

i Q divided by 4, then minus 5

j Q minus 5, then divided by 4



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1.02

2 If $a = 3$ and $b = 2$, evaluate each expression.

a $a + b$ **b** ab **c** $3a + 5b$ **d** $7b - a$

e $8a \div b$ **f** $\frac{a+b}{2}$ **g** $a^2 - b^2$ **h** $2a^2$

1.03

3 Simplify each expression.

a $9y + 7y$ **b** $4ab - 7ab$ **c** $3x^2 + 5x^2$

d $7p + 2q - 5p$ **e** $4w - 5 + 2w$ **f** $8x^2 - 13x - 5x^2$

g $2m + 3n + 5m + 7n$ **h** $5u + 7 - 2u - 3$ **i** $12d^2 + 5d - 7d + 4$

1.04

4 Simplify each expression.

a $9 \times 2n$ **b** $g \times 3h$ **c** $a \times d$

d $(-3) \times 2r$ **e** $12y^2 \div 3y$ **f** $2xy \div y$

g $36a \div 9a$ **h** $15t^2 \div 5t^2$ **i** $20pq \div (-5q)$

1.05

5 Simplify each expression.

a $\frac{h}{6} + \frac{4h}{6}$ **b** $\frac{11k}{8} - \frac{7k}{8}$ **c** $\frac{7}{m} - \frac{5}{m}$

d $\frac{w}{4} - \frac{2w}{5}$ **e** $\frac{4k}{3} - \frac{k}{2}$ **f** $\frac{5}{2d} - \frac{4}{d}$

g $\frac{d}{5k} - \frac{3}{k}$ **h** $\frac{8}{m+1} - \frac{3}{m+1}$

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6 Simplify each expression. 1.06

a $\frac{5}{m} \times \frac{4}{i}$ **b** $\frac{h}{k} \times \frac{3}{k}$ **c** $\frac{14}{3y} \times \frac{9y}{2d}$

d $\frac{r}{4} \div \frac{r}{5}$ **e** $\frac{5}{v} \div \frac{2}{v}$ **f** $\frac{a}{b} \div \frac{a}{b}$

g $\frac{m}{5} \div \frac{5m}{10d}$ **h** $\frac{4}{x} \div \frac{12}{xp}$ **i** $\frac{ad}{6} \times \frac{5a}{2} \div \frac{ad}{8}$

j $\frac{3c}{4d} \div \frac{9c}{16d} \div \frac{6}{8}$ **k** $\frac{15a}{4b} \div \frac{4}{3a^2b} \times \frac{a}{bc}$ **l** $\frac{5y}{8} \div \frac{y}{4} \times \frac{15a}{2y}$

7 Expand and simplify each expression. 1.07

a $5(r - 2)$ **b** $a(a - 10)$ **c** $3y(2x - 5y)$

d $3p(4p - 7q)$ **e** $-6n(n - 9)$ **f** $-6t(t + 7)$

g $b(b + 1) + 5(b + 1)$ **h** $3x(2x - 7) - 5(2x - 7)$ **i** $12 - 6(2x - 3)$

j $5p(2 + 4p - 3q) - (p - q)$ **k** $5(2 - 4p - 3q) - (p + q)$

8 Factorise each expression. 1.08

a $4m - 12$ **b** $xy + xz$ **c** $-3g - 30$

d $16w^2 + 24w$ **e** $-8q + 16$ **f** $15k - 10hk^2$

g $3(y - 6) + y(y - 6)$ **h** $3x(2x - 1) - 2(2x - 1)$ **i** $p(p + 6) - 2(p + 6)$

9 Expand the binomial product $(3d + 5)(d + 4)$ using an area diagram. 1.09

10 Expand each binomial product. 1.09

a $(a + 2)(a + 3)$ **b** $(y + 3)(y - 7)$ **c** $(t - 3)(t + 8)$

d $(h - 5)(h - 4)$ **e** $(4 - g)(3 + g)$ **f** $(3p + 1)(2p + 3)$

g $(5x + 1)(4x - 3)$ **h** $(4r - 5)(3r + 4)$ **i** $(5 - 3q)(2q - 1)$

j $(3m + 5)^2$ **k** $(7 - 2y)^2$ **l** $(3w - 5)(2 - w)$

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Quiz
Test yourself 2

TEST YOURSELF 2

ANSWERS ON P. 566

2.01 **1** Evaluate each expression.

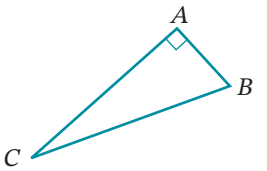
a 8.5^2 **b** $(-6)^2$ **c** $9^2 + 11^2$

2.01 **2** Evaluate each expression correct to 2 decimal places.

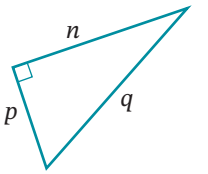
a $\sqrt{146}$ **b** $\sqrt{6^2 + 12^2}$ **c** $\sqrt{27^2 - 16^2}$

2.02 **3** Name the hypotenuse in each triangle.

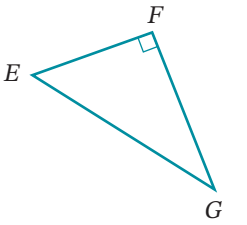
a



b



c

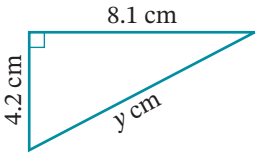


2.02 **4** Write Pythagoras' theorem for each triangle in question **3**.

2.03 **5** Find the value of y. Select the closest answer **A**, **B**, **C** or **D**.

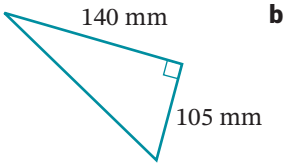
A 12.3 **B** 9.1

C 6.9 **D** 3.5

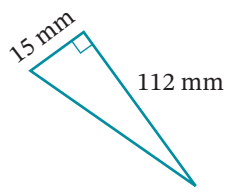


2.03 **6** For each triangle, find the length of the hypotenuse. Give your answer as an exact value.

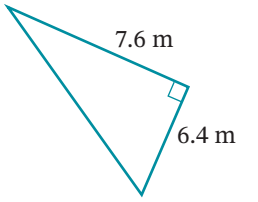
a



b



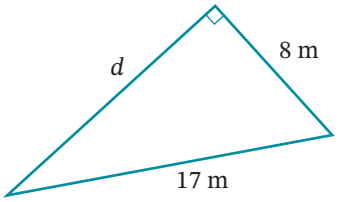
c



2.04 **7** Find the value of d. Select **A**, **B**, **C** or **D**.

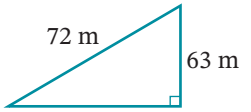
A 12 m **B** 15 m

C 16.8 m **D** 18.9 m

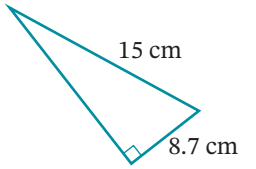


2.04 **8** For each triangle, find the length of the unknown side. Give your answer correct to one decimal place.

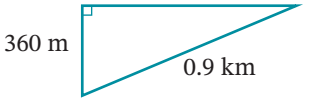
a



b



c

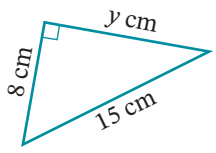


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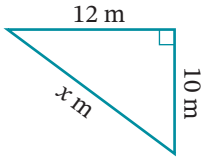
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9 Find the exact value of the variable in each triangle. 2.05

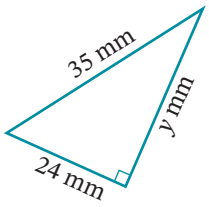
a



b



c



10 In $\triangle DEF$, $\angle F = 90°$, $DF = 84$ cm, and $EF = 1.45$ m. Find DE, correct to the nearest centimetre. 2.05

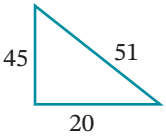
11 Julie walks 6 km due east from a starting point P, while Luka walks 4 km due south from P. 2.05

a Draw a diagram showing this information.

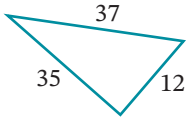
b How far are Julie and Luka apart? Give your answer correct to one decimal place.

12 Test whether each triangle is right-angled. 2.06

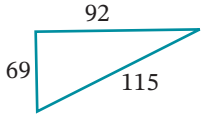
a



b



c



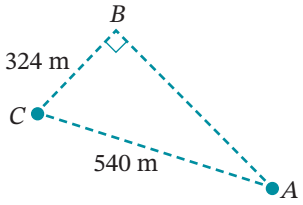
13 Test whether each triad is Pythagorean. 2.07

a {7, 24, 25} **b** {5, 7, 10} **c** {20, 21, 29} **d** {1.1, 6, 6.1}

14 On a cross-country ski course, checkpoints C and A are 540 m apart and checkpoints C and B are 324 m apart. Find the distance: 2.08

a between checkpoints A and B

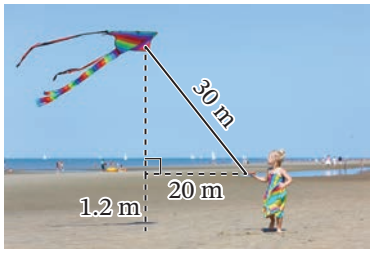
b skied in one lap of the course in kilometres.



15 Miranda holds a kite string 1.2 m above the ground. How high is the kite above the ground? Select **A**, **B**, **C** or **D**. 2.07

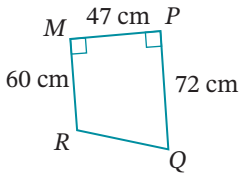
A 22.4 m **B** 23.6 m

C 36.1 m **D** 37.3 m



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16 Find the length of QR in this trapezium, correct to one decimal place. 2.08



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Quiz
Test yourself 3

TEST YOURSELF 3

ANSWERS ON P. 569

3.01 **1** Evaluate each expression.

a $5 \times (-6) \div (-10)$ **b** $8 - (9 + 4)$ **c** $-3 \times 4 - 9$

d $5 \times 3 - 8 \times 5$ **e** $(-5)^2 - 7 \times 3$ **f** $\dfrac{4 \times -8 + (-10)}{3 \times (-2)}$

3.02 **2** Write these decimals in descending order: 0.401, 0.0483, 4.001, 0.05635.

3.02 **3** Evaluate each expression.

a $12 + 0.56$ **b** $14.5 - 6.782$ **c** 4.2×5

d 3.45×0.03 **e** $5.026 \div 4$ **f** $80.45 \div 0.05$

g 7.432×1000 **h** $345.7 \div 100$ **i** $0.605 \div 10$

j $34.1 \times 10\,000$ **k** 856.03×1000 **l** $84.6 \div 100\,000$

3.02 **4** Which expression is NOT the same as $948 \div 4$? Select the correct answer **A**, **B**, **C** or **D**.

A $9.48 \div 0.04$ **B** $94\,800 \div 400$ **C** $0.948 \div 0.004$ **D** $0.948 \div 0.4$

3.02 **5** Round 4.9948 to 3 decimal places.

3.02 **6** Convert $\frac{4}{11}$ to a recurring decimal.

3.03 **7** Which fraction is smaller, $\frac{2}{5}$ or $\frac{5}{12}$?

3.03 **8** Evaluate each expression.

a $\frac{4}{5} + \frac{3}{8}$ **b** $\frac{5}{11} - \frac{2}{3}$ **c** $2\frac{4}{7} - 1\frac{2}{5}$

d $\frac{1}{4} \div \frac{4}{5}$ **e** $\frac{4}{5}$ of \$163 **f** $\frac{1}{2} \times 2\frac{2}{3}$

g $5 \div \frac{3}{4}$ **h** $2\frac{4}{5} \div 1\frac{1}{4}$ **i** $\frac{4}{5} \div \frac{3}{4} - \frac{1}{2}$

3.04 **9** Express each percentage as a simplified fraction.

a 74% **b** 0.5% **c** $66\frac{2}{3}\%$ **d** 14.8%

3.04 **10** Express each percentage as a decimal.

a 21.4% **b** 45% **c** $8\frac{1}{3}\%$ **d** 0.2%

3.04 **11** Express each decimal and fraction as a percentage.

a 0.245 **b** $\frac{7}{8}$ **c** 1.275 **d** $\frac{19}{25}$

3.04 **12** Find:

a 42% of \$580 **b** $8\frac{1}{4}\%$ of 90 kg **c** 2.45% of \$358

3.05 **13** **a** Increase 120 L by 35%.

b Decrease \$25 by 12.5%

3.05 **14** What percentage is (correct to one decimal place):

a 5 minutes of 1 hour?

b 75c out of \$4.50?

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15 In a recent test cricket match, Steve made 286 runs in both innings, which represented 37% of the team total. How many runs did the team make in the match? 3.05

16 Calculate the selling price of a smartphone marked at $485 and discounted by 15%. Select **A**, **B**, **C** or **D**. 3.06

A $72.75 **B** $470 **C** $412.20 **D** $412.25

17 A block of land was purchased for $250 000. Four years later, it was sold for $630 000. Calculate: 3.06

a the profit

b the profit as a percentage of cost price.

18 At a sale, Garrett and Kay bought a cutlery set, saving $150 on the original price. If this discount represented 55% of the price, what was the price of the cutlery? 3.06

19 Calculate, to the nearest cent, the simple interest on: 3.07

a $8050 invested for 5 years at 4.8% p.a.

b $3890 borrowed for 7 months at 2.4% p.a.

20 Calculate the annual simple interest rate if an investment of $7000 earns $1820 in interest over 2 years. 3.07

21 Simplify each ratio. 3.08

a 200 : 450 **b** 35 mm : 2 m **c** 5 h : 45 min

22 Bill and Ben win $800 000 in Lotto. If they agree to share the prize in the ratio of 25 : 17, how much will each person receive? 3.08

23 A truck travels 581 km in 7 hours. 3.08

a Calculate its average speed in km/h.

b How far does the truck travel in 4 hours?

c How long will it take to travel 1000 km? Answer correct to one decimal place.

24 Convert: 3.09

a 0.8 mL/s to mL/h **b** 140 g/m^2 to kg/m^2

c 30 km/h to m/s **d** 345.6 m/day to cm/min

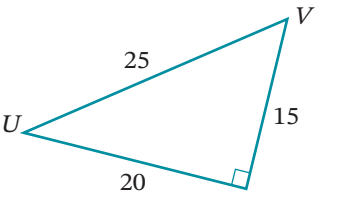
Quiz
Test yourself 4

TEST YOURSELF 4

ANSWERS ON P. 572

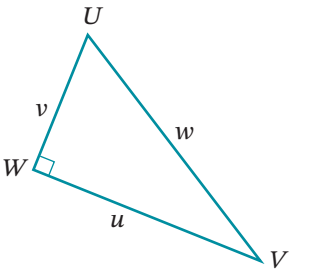
4.01

1 For angle U, name the opposite and adjacent sides and the hypotenuse.



4.01

2 For angle V, name the opposite and adjacent sides and the hypotenuse.

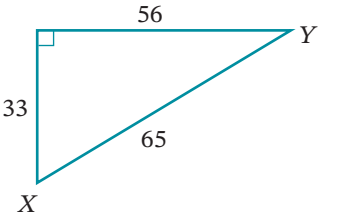


4.02

3 For this triangle, write as a fraction:

a $\sin Y$ **b** $\tan Y$

c $\sin X$ **d** $\cos X$



4.02

4 If $\sin \alpha = \frac{36}{85}$, find the values of $\cos \alpha$ and $\tan \alpha$ as fractions. (Draw a diagram.)

4.03

5 Construct a large right-angled triangle with an angle of 42°, then by measurement and calculation, find the value of each trigonometric ratio, correct to 3 decimal places.

a tan 42° **b** cos 42° **c** sin 42°

Compare your answers to the calculator's answers.

4.04

6 Round each angle to the nearest degree.

a 64° 27′ **b** 25° 43′ **c** 12° 8′ 50″

4.04

7 Round each angle to the nearest minute.

a 50° 19′ 26″ **b** 31° 55′ 55″ **c** 64° 18′ 30″

4.04

8 Evaluate each expression, correct to 4 decimal places.

a cos 32° **b** sin 50° 9′ **c** tan 8° 45′

d 200 tan 18° **e** 14 sin 87° 40′ **f** $\frac{13}{\cos 18°27'}$

4.04

9 Convert each angle size to degrees and minutes, correct to the nearest minute.

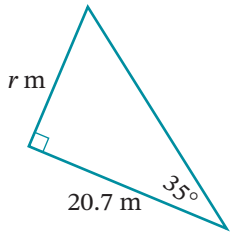
a 45.8° **b** 33.175° **c** 5.346°

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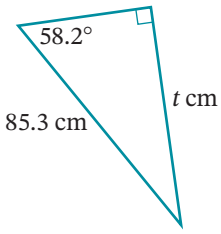
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10 Find the value of each variable, correct to 2 decimal places. 4.05

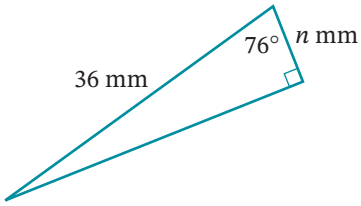
a



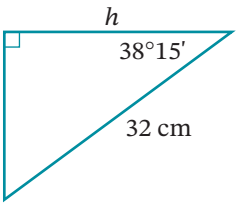
b



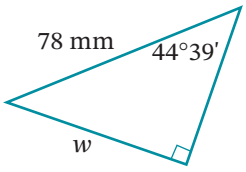
c



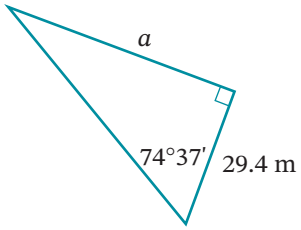
d



e



f

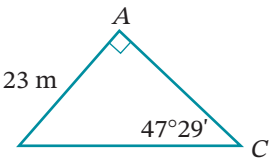


11 For each triangle, find the length of side AC, correct to one decimal place. 4.06

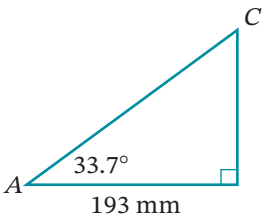
a



b



c

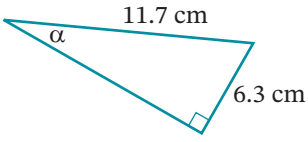


12 Find θ, correct to the nearest degree. 4.07

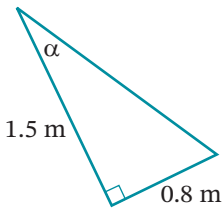
a $\tan\theta = 2.57$ **b** $\cos\theta = \frac{4}{7}$ **c** $\sin\theta = \frac{1.5}{1.6}$

13 Find the size of angle α, correct to the nearest degree. 4.07

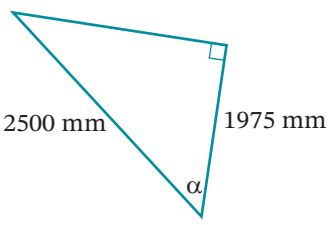
a



b



c



14 In $\triangle AEC$, $\angle C = 90°$, $CE = 3.9$ m and $AE = 4.2$ m. Find $\angle A$, correct to the nearest minute. 4.07

Foundation Standard Complex

Quiz
Test yourself 5

TEST YOURSELF 5

ANSWERS ON P. 575

5.01

1 Simplify each expression, writing the answer in index notation.

a $10^3 \times 10^7$ **b** $4^{20} \div 4^4$ **c** $a^{12} \div a^2$

d $h^8 \times h^2$ **e** $3n^3 \times 4n$ **f** $10d^{15} \div 5d^3$

g $20m^9 \div 4m$ **h** $3v^4w^2 \times 2v^3w^5$ **i** $5x^5y^2 \times 3xy$

j $\frac{24t^8h^8}{3t^4h^2}$ **k** $\frac{p^6q^{10}}{p^2q^2}$ **l** $\frac{100a^2b^4}{5ab^2}$

5.02

2 Simplify each expression, writing the answer in index notation.

a $(2^2)^3$ **b** $(k^5)^5$ **c** $(x^3)^4$

d $(2y^3)^{10}$ **e** $(5t^2)^2$ **f** $(10g)^3$

g $(-2)^5$ **h** $(-2k)^5$ **i** $(-5m^3)^2$

5.03

3 Simplify each expression.

a $(ab^2)^4$ **b** $(5x^3y^2)^2$ **c** $(4t^2)^3$

d $(-4h^2g)^3$ **e** $\left(\frac{a}{7}\right)^4$ **f** $(2pqr)^5$

g $\left(\frac{3m}{2}\right)^5$ **h** $(-3np^2)^4$ **i** $\left(\frac{2a^7}{b}\right)^4$

j $(4t^4u^5)^3 \times 8t^2u$ **k** $\left(\frac{b^8y^6}{8b^2y}\right)^3$ **l** $45c^6d^8 \div (3cd^2)^2$

5.04

4 Simplify each expression.

a 7^0 **b** $(-7)^0$ **c** e^0

d $(-e)^0$ **e** $-e^0$ **f** g^0h

g $(gh)^0$ **h** $\left(\frac{2p}{3}\right)^0$ **i** $\frac{2p^0}{3}$

5.05

5 Simplify each expression using a positive index.

a 8^{-3} **b** 19^{-2} **c** x^{-1} **d** p^{-5}

PATH

5.05

6 Simplify each expression using a positive index.

a $(4m)^{-1}$ **b** $(4m)^{-2}$ **c** $(5b)^{-1}$ **d** $5b^{-1}$

e $-2x^{-4}$ **f** $\left(\frac{3}{5a}\right)^{-1}$ **g** $c^{-4}d^2$ **h** $\left(\frac{100}{9}\right)^{-1}$

5.05

7 Write each expression using a negative index.

a $\frac{1}{10^3}$ **b** $\frac{1}{r^5}$ **c** $\frac{1}{r}$ **d** $\frac{3}{b}$

5.06

8 Simplify each expression.

a $96u^5t^8 \div 24ut^4$ **b** $(5ab)^2$ **c** $4hd^3 \times 5h^3d^6$

d $e^5 \times e \times (e^2)^3$ **e** $(r^3)^4 \div r^2$ **f** $(pn^2)^3 \div (pn^3)^2$

g $\frac{a^8b^3c}{a^4b^3c^2}$ **h** $\frac{60n^4m^3}{12n^8m}$ **i** $\frac{144t^4u^3v}{16u^9v^4}$

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9 Round each value correct to the number of significant figures shown in the brackets. 5.07

a 8.5678 [2] **b** 15 712 [3] **c** 476 [1]
d 0.007 126 6 [4] **e** 0.9041 [3] **f** 301 378 [2]
g 4805.28 [3] **h** 0.000 87 [1] **i** 67 000 000 [1]

10 Express each number in scientific notation. 5.08

a 37 000 **b** 0.61 **c** 250 000
d 0.000 49 **e** 13 **f** 0.000 000 000 08

11 Express each number in decimal form. 5.08

a 8.1×10^{3} **b** 6×10^{7} **c** 3.075×10^{0}
d 8.1×10^{-3} **e** 6×10^{-7} **f** 3.075×10^{-2}

12 Write these numbers in ascending order: 3×10^{3}, 9.1×10^{-8}, 2.4×10^{3}. 5.08

13 Evaluate each expression using scientific notation. 5.09

a $(3.65 \times 10^{22}) \times (7.4 \times 10^{8})$ **b** $(1.44 \times 10^{10}) \div (3.6 \times 10^{4})$
c $(5 \times 10^{5})^{3}$ **d** $\sqrt{6.25 \times 10^{-8}}$

14 Estimate the value of each expression in scientific notation, then evaluate correct to 2 significant figures. 5.09

a $(8.9 \times 10^{9}) \times (1.1 \times 10^{7})$ **b** $(9.3 \times 10^{15}) \times (4 \times 10^{2})$ **c** $(3.1 \times 10^{4})^{2}$

Foundation Standard Complex

Quiz
Test yourself 6

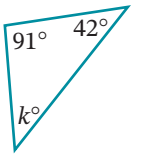
TEST YOURSELF 6

ANSWERS ON P. 578

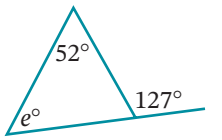
6.01

1 Find the value of each variable, giving reasons.

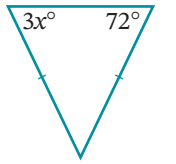
a



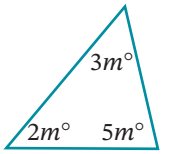
b



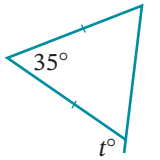
c



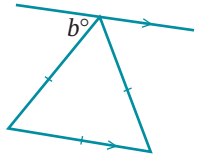
d



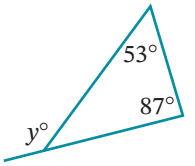
e



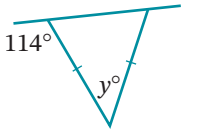
f



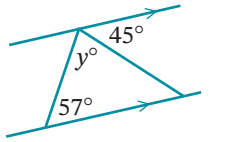
g



h



i



6.01

2 What are the sizes of the angles in:

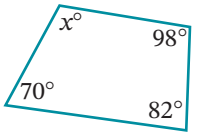
a an equilateral triangle?

b a right-angled isosceles triangle?

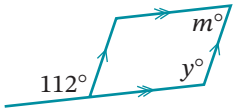
6.02

3 Find the value of each variable, giving reasons.

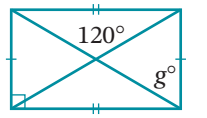
a



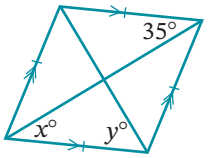
b



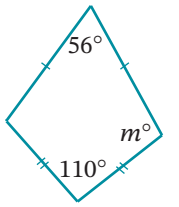
c



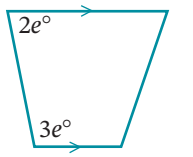
d



e



f



6.02

4 Name all quadrilaterals that have:

a both pairs of opposite sides parallel

b 2 equal diagonals

c all sides equal

d diagonals that bisect each other.

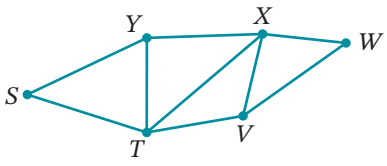
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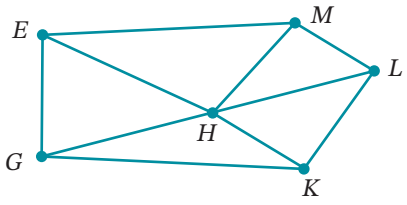
PATH

5 Find the degree of each vertex and show that the sum of the degrees is equal to twice the number of edges. 6.03

a



b



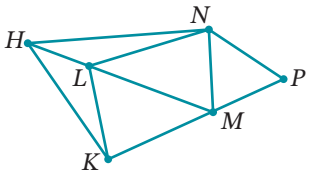
6 **a** State whether each walk is a trail or a path. 6.03

i *HLNPML*

ii *HKMPNLH*

iii *HKMNP*

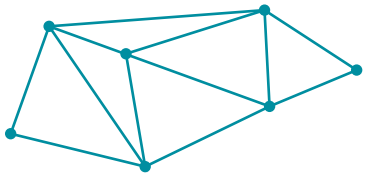
iv *LNMLKM*



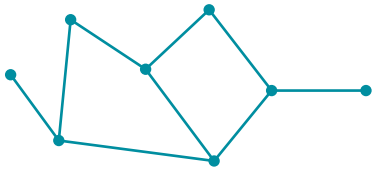
b Which of the walks in part **a** is a cycle?

7 Verify Euler's formula for each planar graph. 6.04

a



b

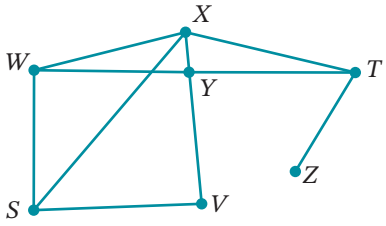


8 Show that each graph is planar by: 6.04

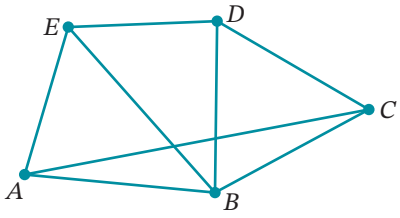
i redrawing it so that no edges cross

ii using Euler's formula

a



b



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TEST YOURSELF 7

ANSWERS ON P. 580

Quiz
Test yourself 7

1 Solve each equation. Check your solutions. 7.01

a $3x - 10 = 5$ **b** $4 + 2y = 21$ **c** $12y + 5 = 23$

d $\frac{m}{4} - 6 = 2$ **e** $\frac{x-5}{4} = 7$ **f** $\frac{5x}{3} = 10$

g $\frac{-2x}{5} = 3$ **h** $11 - 2a = 17$ **i** $20 + 4d = -6$

2 Solve each equation. Check your solutions. 7.02

a $2x + 5 = 3x + 4$ **b** $7 + 4x = x - 8$ **c** $5 - 6w = 3w - 7$

d $3x + 4 = 2x + 7$ **e** $5n - 3 = 2n - 15$ **f** $2d - 8 = -5d - 71$

g $4t = 12 - 4t$ **h** $8j - 17 = 10j$ **i** $6 - 3q = 8 - q$

3 Write an equation with x on both sides that has the solution $x = 3$.

4 Solve each equation. 7.03

a $2(w - 5) = 4$ **b** $3(1 + 4n) = 15$ **c** $5(1 - 3p) = 20$

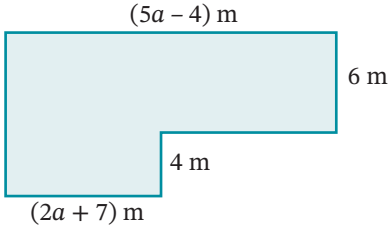
d $2(3 + x) = 5(x + 1)$ **e** $3(1 - y) = 4(2 - y)$ **f** $2(3 - 4x) = -(2x + 3)$

5 Solve each equation. 7.03

a $18 - 3a = 5a - 2(a - 6)$ **b** $11g + 2(g + 3) = 5(2g + 10)$

6 The length of a rectangle is 6 cm longer than it is wide. The perimeter of the rectangle is 76 cm. Find the dimensions of the rectangle. 7.04

7 The diagram shows the floor plan of a house. If the area of the floor is 99 m^2, find the value of a. 7.04



8 If 6 more than a number equals 5 more than double the number, what is the number? 7.04

9 Solve each quadratic equation. 7.05

a $d^2 = 64$ **b** $8p^2 = 288$ **c** $3z^2 = 105$

10 The body mass index (BMI) of an adult is $B = \frac{M}{h^2}$, where M is the mass in kilograms and h is the height in metres. 7.06

a Find the BMI of Ned who is 1.85 m tall and has a mass of 72 kg.

b Find the mass of a person with a BMI of 24, who is 2.1 m tall.

11 The cost, C, in dollars, of hiring a taxi is $C = 5 + 2.4d$, where d is the distance travelled in kilometres. Find: 7.06

a the cost of a taxi trip if the distance travelled is 15 km

b the distance travelled if the cost of a taxi trip was $78.20.

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Quiz
Test yourself 8

TEST YOURSELF 8

ANSWERS ON P. 582

8.01 **1** Milan has an annual salary of $87 410. How much would he be paid:

a per week? **b** per fortnight? **c** per month?

8.01 **2** Diane is paid $23.65 per hour and works a 37-hour week. Find her weekly wage.

8.02 **3** Riley works 38 hours at $37.38 per hour and 5 hours overtime at time-and-a-half. Calculate his total earnings.

8.02 **4** Liam works at a warehouse under the following weekly award:

Award schedule: Normal rate is $24.75 per hour	
Normal rate	For 0 to 38 hours worked
Time-and-a-half	For the next 3 hours worked
Double time	For each hour worked after that

Calculate his wage for working:

a 38 hours **b** 44 hours.

8.03 **5** Nikky is a bookseller who earns 4% commission on the value of books she sells. She is also paid a monthly retainer of $900. If Nicole's sales for March were $82 500, find her total income for March.

8.03 **6** Sam is paid commission on a sliding scale of 3% on the first $50 000 of sales and 1.5% on all sales after that. Calculate Sam's income if he sells goods worth $80 000.

8.03 **7** Calculate the piecework pay for:

a delivering 1500 pamphlets at 50 cents per pamphlet

b washing 45 cars at $27.50 per car

c laying 35.5 m^2 of tiles at $35 per m^2.

8.03 **8** A jockey earns $500 per race and 4.5% of the value of prize money his horses win. Last week, Dan had 12 rides and his horses won prize money worth $120 000. How much did Dan earn?

8.03 **9** Zoe earns $1786 per week and is about to begin her annual holidays. Along with her 4 weeks pay, she also receives a 17.5% annual holiday loading on the 4 weeks. Find Zoe's total holiday pay.

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10 Emily is a department store manager who earned a gross salary last financial year of $187 542 and had the following allowable deductions: $378 uniform cleaning costs, $218 training course costs and $1492 travel expenses. Calculate: 8.04

a Emily's taxable income

b the amount of income tax Emily should pay using this table

Tax rates for Australian residents	
Taxable income	**Tax on this income**
0 – $18 200	Nil
$18 201 – $45 000	19c for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

Source: © Commonwealth of Australia

11 Raj earns $1492.44 per fortnight and pays PAYG tax, $33.15 for life insurance, $20.16 in union fees and $48.17 in superannuation. 8.05

a Use the PAYG table to find Raj's PAYG tax paid per fortnight.

Fortnightly earnings ($)	PAYG tax withheld ($)
1470–1475	174
1476–1481	176
1482–1485	178
1486–1491	180
1492–1497	182

b Calculate Raj's fortnightly net pay.

c Calculate, correct to one decimal place, the percentage of Raj's gross pay that was paid in tax.

12 For the pay slip below, calculate the: 8.05

a gross income **b** total deductions **c** net income.

Employee: Shan Zhou Hours worked		Hourly pay rate: $22.80 Deductions	
Normal	37	Tax: $275.10	Other: $84.50
Time-and-a-half	4	Gross weekly income	
Double time	2	Total deductions	
		Net weekly income	

Foundation Standard Complex

TEST YOURSELF 9

ANSWERS ON P. 586

Quiz
Test yourself 9

1 Find the mean, median, mode and range of this set of data. 9.01

8 6 9 4 10 8 5 5 8

2 A real estate agent sold 10 properties during 1 week. The prices of these properties are: 9.01

$420 000	$485 000	$379 000	$295 000	$1 750 000
$515 500	$315 500	$408 000	$368 000	$315 000

a Find the mean price of the properties.

b Find the median price of the properties.

c Compare the mean price with the median price. Which is the better measure of centre? Give reasons.

d When data on sales of properties are given, the median price is used, not the mean. Explain why.

3 These percentage marks were scored by 30 students in a Maths test. 9.02

45	67	78	54	22	38	58	64	73	89
84	70	78	62	56	59	42	47	86	90
64	63	72	77	43	85	75	71	70	83

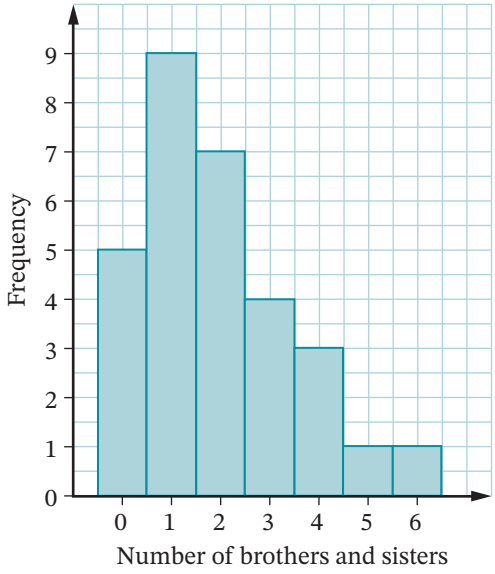
a Construct an ordered stem-and-leaf plot for the data.

b Find the mean, median, mode and range of the marks.

c Are there any outliers?

d Compare the mean and median. Which is more accurate? Give reasons.

4 A group of people were surveyed on the number of brothers and sisters they had. The results are displayed on the histogram. 9.02



a What is the range of this data?

b What is the mode?

c Find the mean correct to 2 decimal places.

d What is the median?

e Which is the best measure of centre?

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9.02

5 24 patients were chosen at random from 2 medical centres. The waiting times (in minutes) for each patient are listed below.

Northbridge:	7	16	28	27	30	30	47	9	9	15	27	11
	38	35	42	53	23	22	45	32	37	37	18	16
Westvale:	44	13	7	9	25	26	37	35	8	18	18	15
	22	24	7	5	24	24	33	37	24	15	36	43

a Display the information in a back-to-back stem-and-leaf plot.

b Find the mean, median and range of both sets of data.

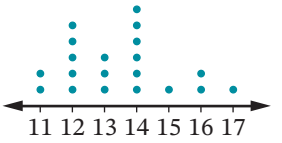
c Identify any outliers or clusters.

d Are there significant differences between the 2 sets of data? Explain your answer.

9.03

6 For each distribution, describe the shape and identify any outliers or clusters.

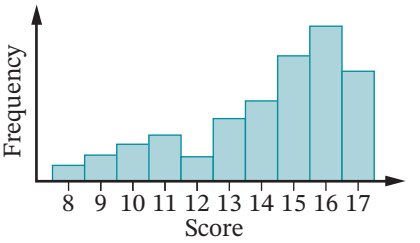
a



b

Stem	Leaf
8	7 8
9	1 3 4 4
10	0 2 7 8 8 9
11	1 2 2
12	0 7

c



9.04

7 The heights (in centimetres) of students in a Year 9 class are:

Girls:	160	153	155	156	142	163	162
	148	150	150	131	138	147	145
Boys:	150	165	164	178	170	171	164
	148	154	176	134	180	176	175

a Arrange the data in a back-to-back stem-and-leaf plot.

b Describe the shape of the display for:

i the girls **ii** the boys.

c Are there any outliers or clusters for either group?

d Find the mean (correct to 2 decimal places) and median for both groups.

e Which measure is the better indicator of the measure of centre?

f Which group is shorter?

g Which group has the greater spread of heights?

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8 Should a sample or a census be used to investigate each topic?

PATH 9.05

- **a** The most popular car colour
- **b** Whether energy-saving light globes really use less electricity
- **c** The number of retired people living on the Gold Coast, Queensland
- **d** People's views on which nation will win the soccer World Cup

9 Classify each type of data as categorical or numerical. If categorical, then classify it as nominal or ordinal. If numerical, then classify it as discrete or continuous.

9.05

- **a** The temperature in Alice Springs.
- **b** The number of views of a TikTok video.
- **c** Suburb or town of home.
- **d** The make of car passing through a motorway toll collection point.
- **e** The probability of rain today.

10 Explain why each of the following samples is biased.

PATH 9.06

- **a** A sample of online gamers were surveyed to investigate teenagers' views on road safety.
- **b** Customers in a café can give feedback about their meal by filling in a survey and posting it to the café.
- **c** At a factory, a sample of executives were surveyed about safety in the workplace.
- **d** A sample of audience members at a movie premiere were surveyed about their views on the film.
- **e** Classic Gold FM radio listeners were asked to vote for the best song of the year.

11 Explain what is wrong with each survey question and suggest an alternative.

9.06

- **a** Are you happy?
- **b** How did you enjoy your meal at our restaurant?
- **c** Australian soldiers have fought under our country's flag. Do you think that it should be changed?

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TEST YOURSELF 10

ANSWERS ON P. 590

1 Convert: 10.01

a 8 km to m **b** 15 min to s **c** 0.7 m to mm
d 250 000 g to kg **e** 8.4 GB to MB **f** 9.5 t to kg
g 300 min to h **h** 34 000 B to kB **i** 125 kL to L
j 2500 cm to km **k** 17 000 000 kg to t **l** 8900 GB to TB
m 10 500 ms to s **n** 9.1 millennia to years **o** 2.6 Mm to m
p 3 ly to km **q** 0.000 000 4 s to μs **r** 75 000 000 μm to m

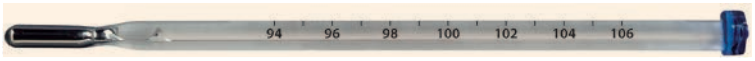
2 For each measuring instrument, state: 10.02

i its precision **ii** its absolute error.

a



b

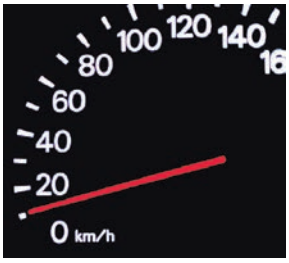


c



Shutterstock.com/Sergey Melnikov

d

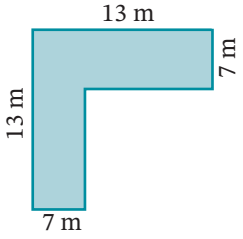


3 A super yacht is 24 m long. For this measurement, calculate: 10.03

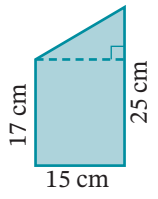
a its absolute error **b** its percentage error correct to 2 decimal places

4 Calculate the perimeter of each shape. 10.04

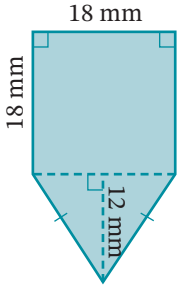
a



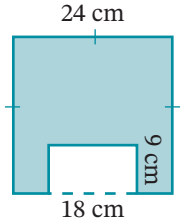
b



c



d

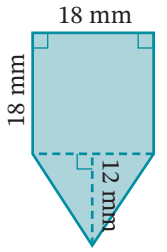


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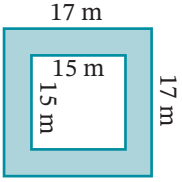
10.04

5 Calculate the area of each shape.

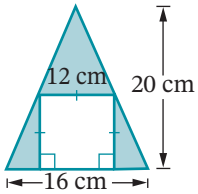
a



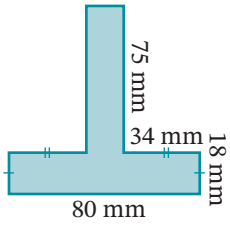
b



c



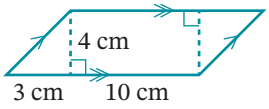
d



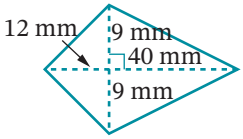
10.05

6 Calculate the area of each quadrilateral.

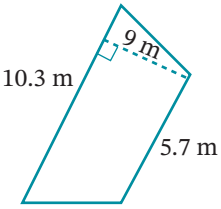
a



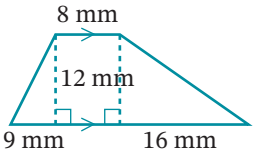
b



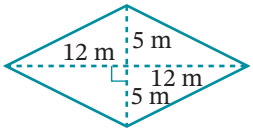
c



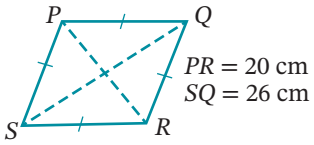
d



e



f

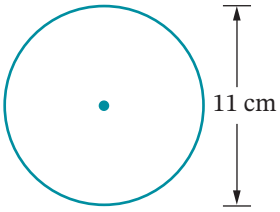


10.06

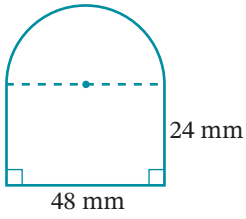
7 For each shape calculate, correct to one decimal place:

i its perimeter **ii** its area.

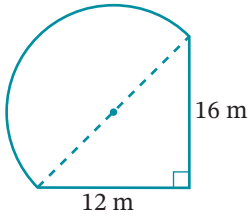
a



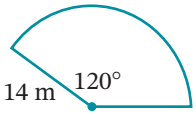
b



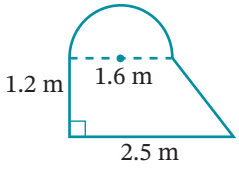
c



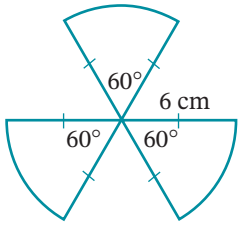
d



e



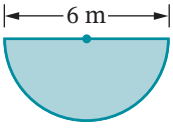
f



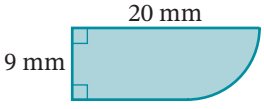
10.06

8 Calculate, correct to 2 decimal places, the area of each shape.

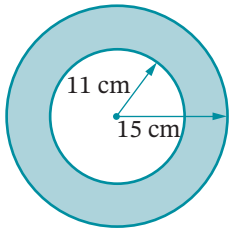
a



b



c

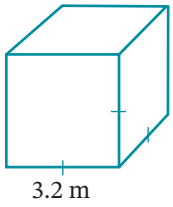


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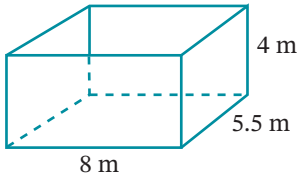
9780170479370

9 Calculate the surface area of each prism. 10.07

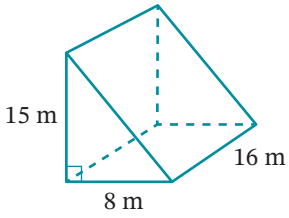
a closed cube



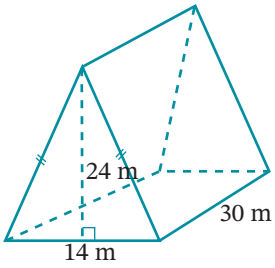
b open rectangular prism



c



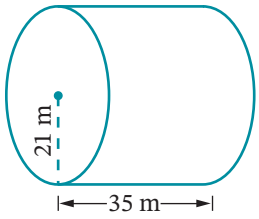
d



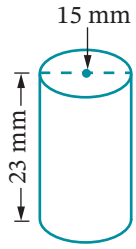
10 Calculate the volume of each prism in question **8**. 10.08

11 Calculate, correct to 2 decimal places, the surface area of each cylinder. 10.08

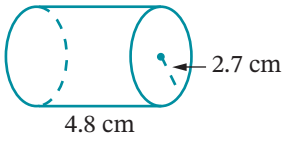
a



b



c



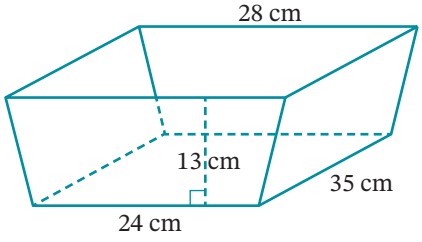
Cylinder, open at one end

12 Calculate, correct to 2 decimal places, the volume of each cylinder in question **10**. 10.09

13 A cylindrical pipe (open at both ends) is 50 cm long and has a diameter of 15 cm. Calculate its external surface area correct to the nearest square centimetre. 10.08

14 A rectangular fish tank that is 75 cm long by 55 cm wide by 35 cm deep is filled to 5 cm from the top. How many litres of water is in the tank? 10.09

15 Calculate the volume of this lunchbox. 10.09



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TEST YOURSELF 11

ANSWERS ON P. 601

Quiz
Test yourself 11

1 Find the length of the interval joining the points $H(7, 2)$ and $T(-2, 6)$. Express your answer: 11.01

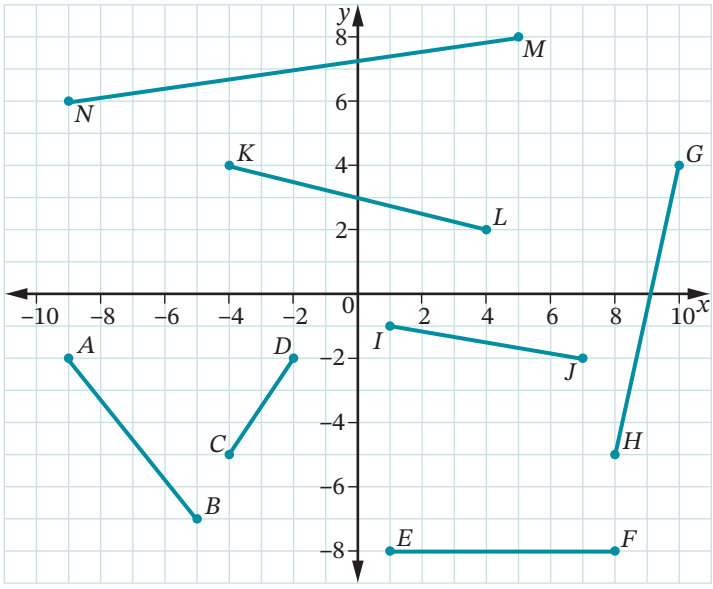
a as a surd **b** correct to one decimal place.

2 Find the coordinates of the midpoint of the interval joining: 11.02

a (8, 4) and (−2, 10) **b** (−1, −1) and (−7, 9) **c** (3, −6) and (−5, 10)

3 **a** Which intervals below have a gradient that is: 11.03

i positive? **ii** negative? **iii** 0?



b Find the gradient of:

i interval NM **ii** interval AB **iii** interval KL

c Find the midpoint of interval HG.

d Find the exact length of interval CD.

4 Find the gradient of the interval joining: 11.03

a (1, 9) and (3, 5) **b** (6, 0) and (8, −4) **c** (−1, −1) and (5, −3)

5 Graph $y = -3x + 6$ on a number plane and write: 11.04

a its x-intercept **b** its y-intercept.

6 Test whether the point (4, −2) lies on the line with equation: 11.04

a $y = x - 6$ **b** $y = -2x + 6$ **c** $x - 3y = 10$

7 **a** Graph and label each line on the same number plane: 11.04

$x = -4$, $x = 3$, $y = -6$, $y = 2$

b What is the area of the rectangle enclosed by these lines?

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11.04

8 Write the equation of a line that is:

a horizontal with a y-intercept of 4

b parallel to the y-axis and passes through $(-1, 7)$

c the x-axis

11.05

9 Find the gradient and y-intercept of the line with equation:

a $y = 3x + 1$ **b** $y = 8 - x$ **c** $y = \frac{x}{5} + 4$ **d** $y = -6x$

11.05

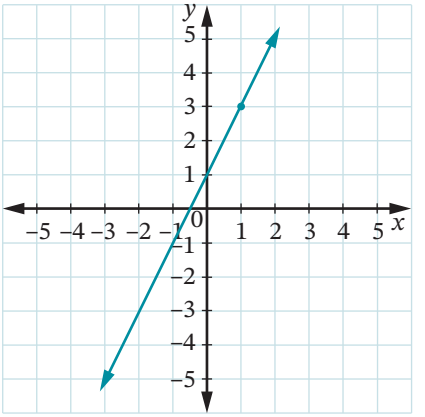
10 Graph each linear equation by finding the gradient and y-intercept first.

a $y = -3x - 1$ **b** $y = \frac{x}{2} + 6$

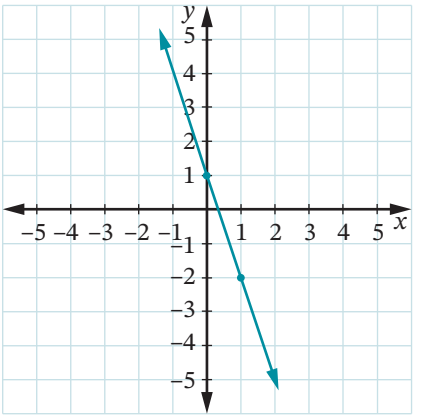
11.06

11 Find the equation of each line.

a



b



11.07

12 Solve the equation $3x - 4 = 5$ graphically.

11.08

13 The cost of using a ridesharing service is a flagfall of \$3.50 and then a charge of \$2.25 per kilometre travelled.

a Copy and complete this table.

Distance (x km)	0	1	2	3	4	5	6
Cost (\$$y$)							

b Graph the table of values.

c Use the graph to find the cost of a 3.5 km trip.

d Find the gradient and y-intercept of the line.

e Find the equation of the line.

f Use the equation to find the cost of the taxi for a distance of:

i 12 km **ii** 76 km

g Find the distance travelled if the cost was:

i \$12.05 **ii** \$82.25

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PATH

14 During a thunderstorm, the time, T seconds, between seeing lightning and hearing thunder varies directly as the distance, d kilometres, of the observer from the centre of the storm. Thunder is heard 6 seconds after the lightning is seen when a storm is 35 km away. 11.09

a Find a formula for T in terms of d.

b Calculate, correct to the nearest kilometre, the distance of a storm when thunder is heard 10 seconds after the lightning.

15 Graph $y = x^2 + 1$ and $y = -2x^2$ on the same set of axes and state: 11.10

a the vertex of each parabola

b whether each parabola is concave up or concave down

c which parabola is 'wider'.

16 Which equation below represents a concave up parabola with vertex (0, 0)? Select the correct answer **A**, **B**, **C** or **D**. 11.10

A $y = x^2 - 1$ **B** $y = 2x^2$ **C** $y = -x^2$ **D** $y = 3x^2 + 2$

17 A stone is dropped from a cliff and its height y metres at time t seconds is given by $y = -5t^2 + 20$. 11.11

a Copy and complete this table of values and graph the equation.

Time (t seconds)	0	1	2	3
Height (y metres)				

b What is the y-intercept and what does it represent?

c What is the height of the stone after:

i 0.5 seconds? **ii** 1.5 seconds?

d When will the height of the stone be 5 m?

e When will the stone hit the ground?

f What distance does the stone drop in the first second?



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Quiz
Test yourself 12

TEST YOURSELF 12

ANSWERS ON P. 605

12.01

1 A jar contains only red and yellow marbles. The probability of selecting a red marble from the jar is 40%.

a What is the probability of selecting a yellow marble?

b If there are 40 marbles in the bag, how many red marbles are there?

12.01

2 A card is drawn randomly from a normal deck of cards. Find the probability that it is:

a a red card

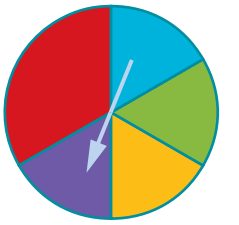
b a King or a Queen card

c a card with a number greater than 6.

12.02

3 For the spinner shown, the red sector is twice as large as each of the other sectors. The spinner was spun 200 times and the results are shown in the table.

Outcome	Frequency
Blue	29
Green	35
Yellow	30
Purple	34
Red	72



a Find the observed probability (as a decimal) that the arrow lands on:

i red **ii** blue or green **iii** not purple **iv** green.

b Find the theoretical probability (as a decimal) that the arrow lands on:

i red **ii** blue or green **iii** not purple **iv** green.

c Are the observed probabilities from part **a** similar to the theoretical probabilities from part **b**?

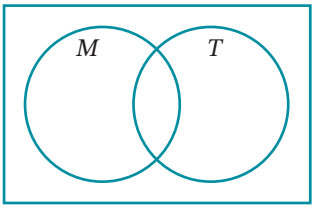
d If the spinner is spun 500 times, calculate the expected frequency of spinning red using the theoretical probability. Select the correct answer **A**, **B**, **C** or **D**.

A 100 **B** 85 **C** 167 **D** 200

PATH

12.03

4 30 teenagers were surveyed on whether they preferred to listen to music (M) or watch TV (T). 21 students preferred music and 18 liked TV, while 4 students did not like either.



a How many students preferred both music and TV?

b Copy and complete the Venn diagram.

c Find the probability of randomly selecting a teenager who likes:

i music but not TV **ii** TV

iii music or TV, but not both **iv** both or neither music nor TV.

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PATH

12.04

5 A group of men and women was surveyed on whether they worked full-time or part-time.

	Full-time work	Part-time work
Male	56	41
Female	45	68

a How many people were surveyed?

b Find the probability (as a percentage, correct to one decimal place) of selecting a person who:

i works full-time

ii is female and works part-time

iii does not work full-time

iv works part-time or is female.

c If a part-time worker is selected at random, what is the probability that the person is female?

12.05

6 4 cards labelled with the letters A, B, C and D are placed in a bag.



2 cards are drawn from the bag at random without replacement.

a List all of the possible outcomes.

b Copy and complete the tree diagram.

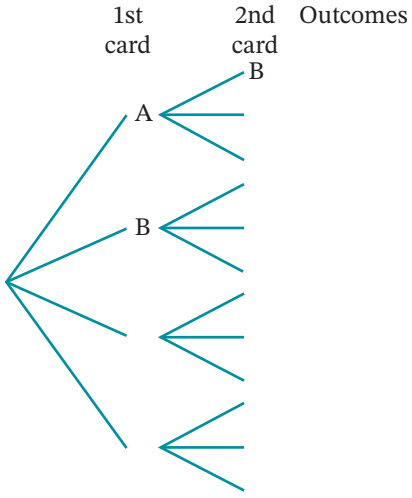
c Use the tree diagram to find the probability of drawing 2 cards with:

i the same letter

ii the letters A and B

iii the letters A or B or both

iv no letter D.



12.06

7 2 dice are rolled and the difference between the numbers is calculated.



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Die 2 \ Die 1	1	2	3	4	5	6
1						
2						
3						3
4						
5				1		
6		4				

6 − 3 = 3

5 − 4 = 1

6 − 2 = 4

a Copy and complete the table to show the sample space.

b Find the probability of rolling a difference of:

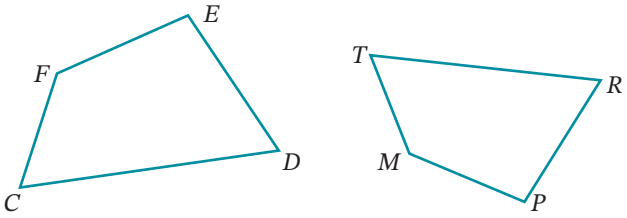
i 0 **ii** at least 2 **iii** 4 or 5.

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TEST YOURSELF 13

ANSWERS ON P. 608

1 The quadrilaterals *FECD* and *TRMP* are similar. 13.01



a List all pairs of matching sides.

b List all pairs of matching angles.

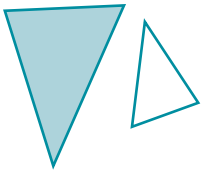
c Use the correct notation to write a congruency statement relating the 2 quadrilaterals.

2 By measurement, find the scale factor for each pair of similar figures. 13.01

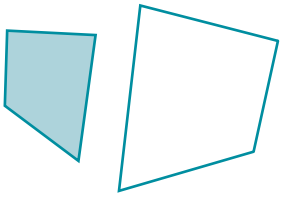
a



b



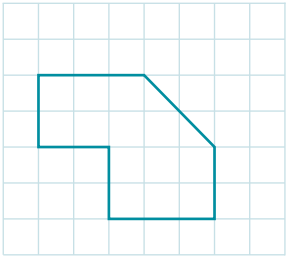
c



3 Copy each figure onto graph paper and draw its image if: 13.01

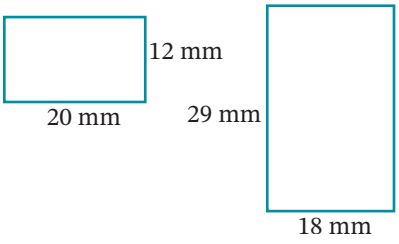
a scale factor = 3

b scale factor = $\frac{1}{2}$

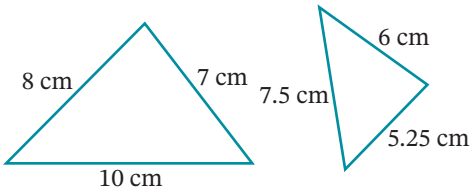


4 Test whether each pair of figures are similar. 13.02

a



b



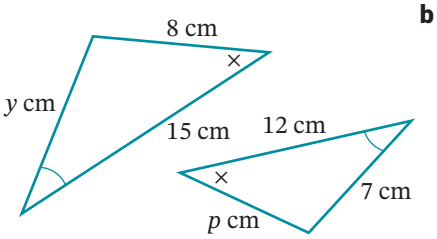
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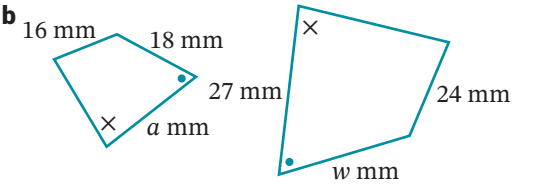
13.02

5 Find the value of each variable in each pair of similar figures.

a



b



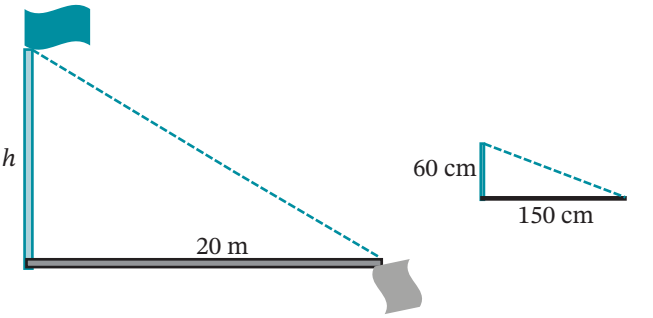
13.03

6 A flagpole's shadow is 20 m long. At the same time, a stick 60 cm tall casts a shadow 150 cm long.

What is the height, h, of the flagpole? Select **A**, **B**, **C** or **D**.

A 12 m **B** 50 m

C 8 m **D** 9 m

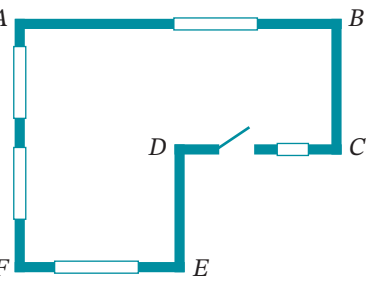


13.03

7 This floor plan of a holiday house is drawn to a scale of 1 : 200. What is the actual (internal) length of:

a AB?

b CD?



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